CHAEEUN KIM

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Researcher | Seoul National University | Republic of Korea

Research Interests | *Biosensors; Bioelectronics; Wearable sensing; Point-of-care Diagnostics; Biosensor-integrated organ-on-chip systems*

Education

Seoul National University, Seoul, Republic of Korea Feb 2026

- M.S. in Applied Bioengineering. Advisor: Prof. Kangwon Lee.
- Thesis: Calibration-Free Potassium Quantification via a Stoichiometric Enzyme Cascade on Laser-Induced Graphene **(Outstanding Thesis Award)**

Chung-Ang University, Seoul, Republic of Korea Feb 2024

- B.S. in Biomedical Engineering.
- Thesis: Trends in Antigen-Antibody-Based Biosensor Technology for Pathogen Diagnosis.

Research Experience

Advanced Functional Biomaterials Laboratory, Seoul National University Jul 2023 – Present
(*Advisor: Prof. Kangwon Lee)*
Researcher Mar 2026 – Present

- Advanced the commercialization of a potassium-ion sensing platform through device integration, performance validation, and coordination with external testing institutions.
- Fabricated electrospun components for a MOTIE-funded 3D cell-based regenerative therapy project, supporting the development of materials and devices for cell preservation and thawing.

Graduate Researcher Mar 2024 – Feb 2026

- Developed a calibration-free potassium sensor using a pyruvate kinase/pyruvate oxidase cascade on laser-induced graphene; led experimental design, optimization, and first-author manuscript preparation.
- Achieved a 2–8 mM linear range and inter-sensor RSD of 2.41–5.20%; validated performance in potassium-spiked human serum.
- Contributed to the clinical evaluation of disposable potentiometric potassium-sensing strips using capillary blood from patients receiving maintenance hemodialysis in collaboration with Severance Hospital; co-first author of the clinical study.
- Integrated carbon-coated δ-MnO_2 nanosheets into potassium-sensing strips with Ewha Womans University, contributing to a 1.4-fold faster response than that of activated-carbon solid contacts.

Undergraduate Research Intern Jul 2023 – Feb 2024

- Worked on the development of potassium ion-selective electrodes, including electrochemical sensor fabrication, characterization, and optimization.

Translational Multiscale Biofluidic Laboratory, UNIST *(Advisor: Prof. Joo Hun Kang)* Jul 2022
Undergraduate Research Intern (U-SURF program)

- Trained in magnetic blood cleansing, μFISH-based bacterial identification, and vascularized microfluidic networks.

Department of Preventive Medicine, Seoul National University College of Medicine Jun 2021 – Aug 2021
(Advisor: Prof. Aesun Shin)
Undergraduate Research Intern

- Reviewed literature on genetic variants associated with colorectal cancer susceptibility in Asian populations.

Publications

1. **Chaeeun Kim**, Chaehyung Kim, Yujoon Kim, and Kangwon Lee*. Calibration-free potassium quantification via a stoichiometric enzyme cascade on laser-induced graphene. *Biosensors and Bioelectronics* 304, 118572 (2026). DOI: 10.1016/j.bios.2026.118572

2. Cheol Ho Park†, Yujoon Kim†, **Chaeeun Kim**†, Chaehyung Kim†, Sowon Kim, Kangwon Lee, and Tae-Hyun Yoo*. Clinical evaluation of a disposable test strip for potassium level measurement in patients treated with maintenance hemodialysis. *Clinical Journal of the American Society of Nephrology* 21(3), 425–436 (2026). DOI: 10.2215/CJN.0000000934, †Equal contribution.

3. Jaeyeon Lee, **Chaeeun Kim**, Chaehyung Kim, Yujoon Kim, Seojin Lee, Kangwon Lee*, and In Young Kim*. Supercapacitive carbon-coated δ-MnO_2 nanosheets as a solid-contact material for self-testing potassium-ion biosensors. *Small* 22(26), e13386 (2026). DOI: 10.1002/smll.202513386

Patents and Intellectual Property

1. **Strips for measuring glucose and potassium** Jan 2025
   Yujoon Kim, **Chaeeun Kim**, et al. Korean Patent No. 10-2757781 (granted).

Conference Presentations

1. **Chaeeun Kim**, Kangwon Lee*, "Long-Term Storage-Stable Potassium Sensing Enabled by a PVB-Enhanced Reference Electrode." SNU BioDay – Poster presentation Sep 2025
2. **Chaeeun Kim**, Chaehyung Kim, Kangwon Lee*, "Polyvinyl Butyral (PVB)-Enhanced Reference Electrode for Stable Potassium Sensing in Point-of-Care (POC) Applications." NANO KOREA – Poster presentation Jul 2025
3. **Chaeeun Kim**, Chaehyung Kim, Yujoon Kim, Kangwon Lee*, "Polyvinyl Butyral (PVB)-Enhanced Reference Electrode for Stable Potassium Sensing in Point-of-Care (POC) Applications." Korean BioChip Society – Poster presentation **(Best Poster Presentation Award)** Nov 2024

Selected Projects

Portable Potassium-Ion Measurement Device Development, THEDONEE Inc. Jul 2023 – Present

- Led translational R&D for a portable potassium-ion measurement device and disposable test-strip platform.
- Coordinated sensor validation, measurement-circuit and firmware development, device housing design, and collaboration with external testing institutions.
- Contributed to prototype planning, performance verification, and clinical-use-oriented product development.

Simultaneous Glucose and Potassium Test Strip, Seoul R&D Program Sep 2023 – Aug 2025

- Seoul Business Agency–funded project BT230224; led potassium-sensor experimental design, data analysis, and system integration as a student researcher; coordinated material–device research with Ewha Womans University.

Honors and Scholarships

Outstanding Thesis Award, *Graduate School of Convergence Science and Technology, Seoul National University*	Feb 2026
Lecture & Research Scholarship, *Seoul National University*	Fall 2025
Best Poster Presentation Award, *Korean BioChip Society*	Nov 2024
Merit-based Scholarship, *Seoul National University*	Fall 2024
Audience Choice Award, ICC CORE-ATHON Startup Competition	Dec 2022
Scholarship, *Ulsan Institute for Talent and Lifelong Education*	Fall 2021

Research & Technical Skills

Electrochemical Methods: Chronoamperometry (CA), Cyclic Voltammetry (CV), Potentiometry

Materials Characterization: Scanning Electron Microscopy (SEM), X-ray Diffraction (XRD), Raman Spectroscopy

Fabrication & Assay Development: Laser-Induced Graphene (LIG) Fabrication, O_2 Plasma Treatment, Enzyme-Cascade Optimization, Ion-Selective Electrode (ISE) Fabrication, Electrospinning

Optical Analysis: UV–Vis Spectrophotometry, Microplate-Based Absorbance Assays

Analytical & Clinical Validation: Sensor Calibration, Selectivity and Reproducibility Testing, Clinical Sample Analysis

Software: R, OriginPro, GraphPad Prism, ImageJ, AutoCAD